

ESPRIT · 2026

FINAL YEAR PROJECT

PEAXIS

A Modular AI-Powered Hiring Operating System

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1 INTRODUCTION & CONTEXT

Presentation Overview

01 Context & Problem

02 Solution & Demo

03 Architecture

04 AI Engineering

05 Verification & Roadmap

06 Conclusion

2 INDUSTRY & PROBLEM

Recruitment in the Digital Era



Application volume



Recruiter workload



Candidate expectations



AI adoption in HR

2 COMPANY PRESENTATION

Prospecter — Internship Host Company

AI-powered B2B platform for outbound prospecting.

MISSION

Autonomous AI SDRs help B2B teams scale outbound prospecting.

AI / LLM

B2B SAAS

SALES AUTOMATION

OUTBOUND GTM



AI SDR Engine



Lead Intelligence



Outbound Automation



Pipeline Analytics



Strategic context — LLM workflows · Multi-tenancy · Background workers

3 INTERNSHIP

Internship Context

Software Engineering internship at **Prospecter**, an AI-powered B2B platform.



Full-Stack Engineering



AI Integration



Platform Engineering

Scope: Independent product · Production-inspired engineering

SOFTWARE ENGINEER

AI INTEGRATION

SOLO CONTRIBUTOR

2 INDUSTRY & PROBLEM

Current recruitment challenges

42 days

Average time-to-hire
SHRM 2024

75%

Recruiters overloaded
LinkedIn Talent Report

60%

Candidates report no feedback
Indeed Survey 2023



Manual screening



Keyword-based ATS



Inconsistent evaluation



Candidate experience

Cost of a bad hire: averages **\$14.9K**; strong candidates can accept competing offers within days.

Engineering response: keep screening responsive, make every score reviewable, and keep candidates informed.

4 EXISTING SOLUTIONS & GAP

The gap: **integrated, explainable AI**



AI black boxes



Fragmented tooling



No candidate intelligence

Evidence first

Human authority

Durable operations



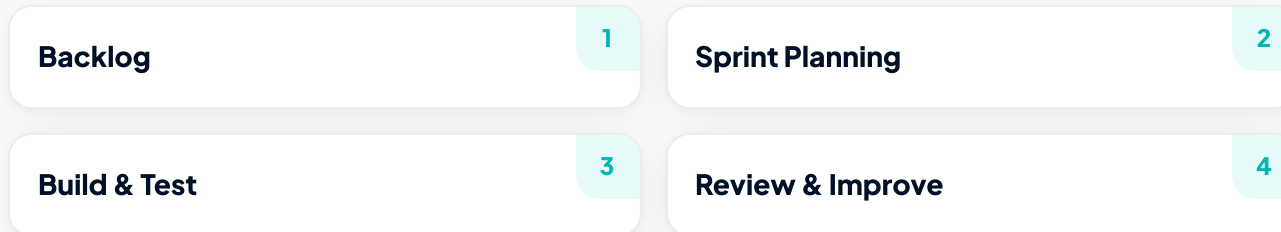
This gap defines the project's objective

One auditable recruitment platform.

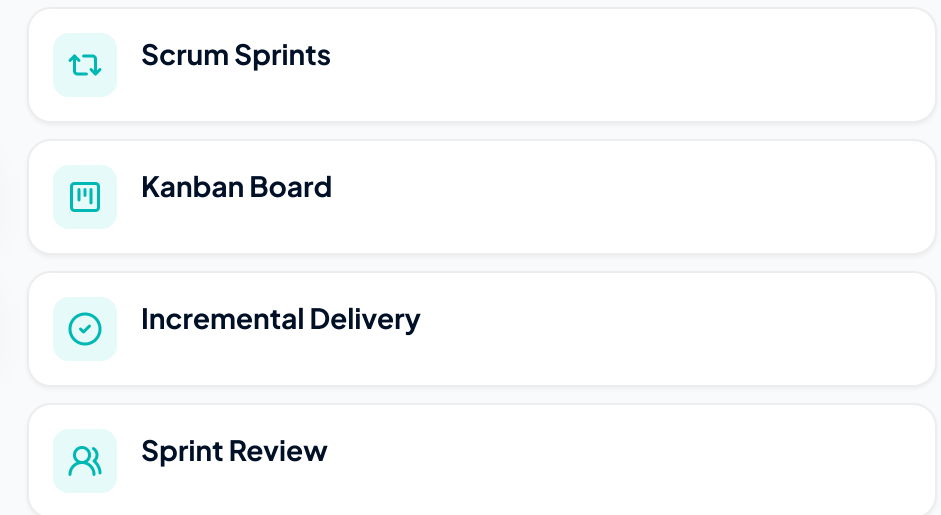
5 METHODOLOGY & REQUIREMENTS

Engineering Methodology

SPRINT CYCLE



HOW WORK WAS MANAGED



5 FUNCTIONAL REQUIREMENTS

Core platform requirements

Core platform capabilities.



FR-01

Authentication & Security

- Argon2 password protection
- JWT + OAuth sign-in



FR-02

Multi-Tenant Architecture

- Business-scoped records
- Tenant context enforced



FR-03

Roles & Permissions

- Role and plan guards
- Feature access by subscription



FR-04

Job Management

- Create and filter jobs
- Hybrid text + vector search



FR-05

Application Pipeline

- Tracked application stages
- Async assessment updates

5 FUNCTIONAL REQUIREMENTS

AI requirements

Implemented intelligence capabilities.



FR-06

Evidence Matching

- Requirement-level assessment
- 0-100 alignment score



FR-07

CV Parsing

- PDF / DOCX / TXT upload
- Async structured extraction



FR-08

Candidate Onboarding

- Apply with a CV
- Confirm parsed profile



FR-09

Explainable Review

- Cited strengths and gaps
- Verification state per requirement



FR-10

Reliable AI Processing

- Queued AI work items
- Retry and failure status

5 NON-FUNCTIONAL REQUIREMENTS

Quality attributes

Reliability, security, and maintainability.



Container-ready

NFR-01 **Deployment Readiness**

Web, API, worker, AI runtime, and data services are separated.



Implemented controls

NFR-02 **Security**

Auth, file gates, rate limits, tenant scoping, and service secrets.



No AI in request path

NFR-03 **Responsive Requests**

Parsing and assessment run outside the HTTP request path.



Explicit work states

NFR-04 **Failure Handling**

Retries and durable work states expose processing failures.



Cited assessment trail

NFR-05 **Explainability**

Scores retain evidence, gaps, and verification state.



Modular platform

NFR-06 **Modularity**

Separate web apps, API, worker, and inference boundary.

6 PROPOSED SOLUTION

PEAXIS — Modular AI Hiring Operating System



Business foundation

PEAXIS Core



Recruiter workspace

PEAXIS Hire



Candidate portal

PEAXIS Jobs

BUSINESS ADMIN

PEAXIS Core

RECRUITER / HR TEAM

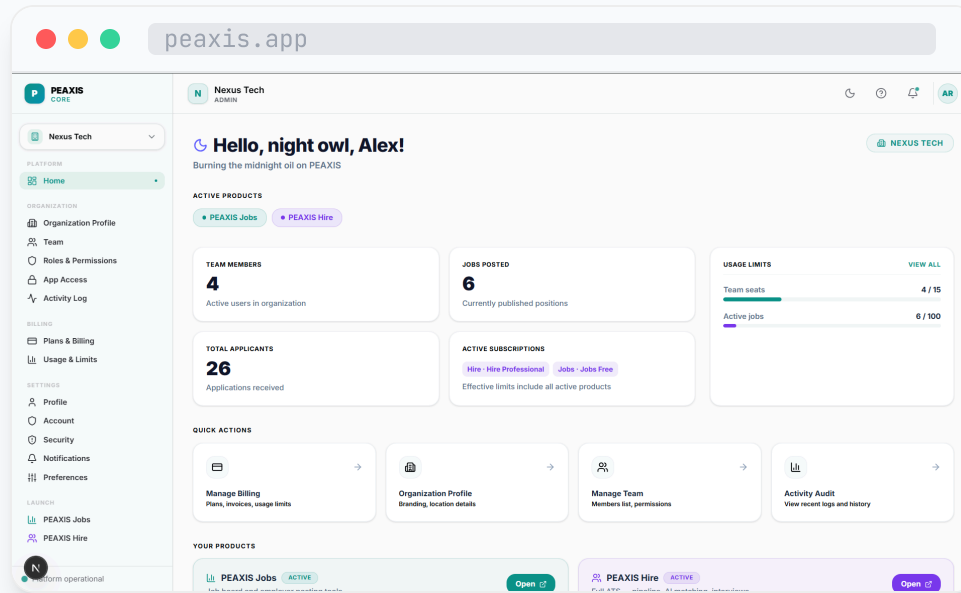
PEAXIS Hire

CANDIDATE / TALENT

PEAXIS Jobs

6 PROPOSED SOLUTION

PEAXIS Core



PEAXIS Core — access, business context, and plan controls



Identity & Access



Plans & Entitlements

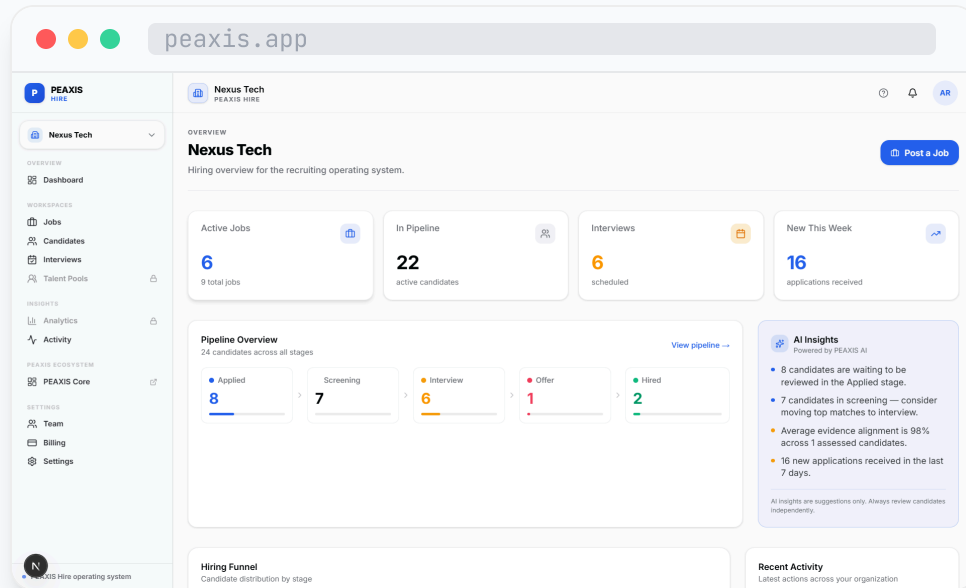


Business Context

One control centre for every module

6 PROPOSED SOLUTION

PEAXIS Hire



PEAXIS Hire — recruiter pipeline and assessment workspace

 **Pipeline workspace**

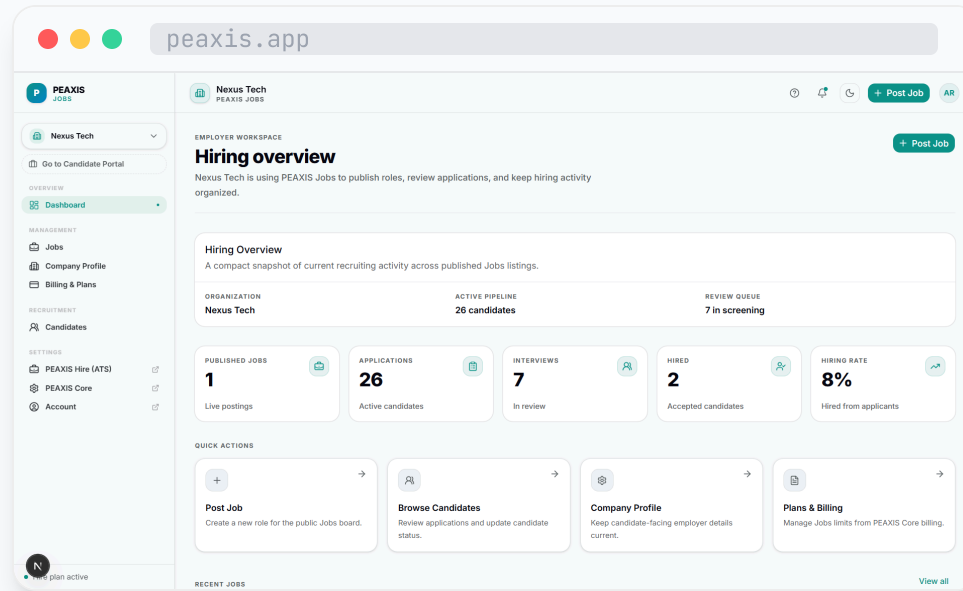
 **Evidence assessment**

 **Async processing**

 Pipeline → evidence → recruiter decision

6 PROPOSED SOLUTION

PEAXIS Jobs



PEAXIS Jobs — job discovery and candidate application portal



Job Discovery



Application UX



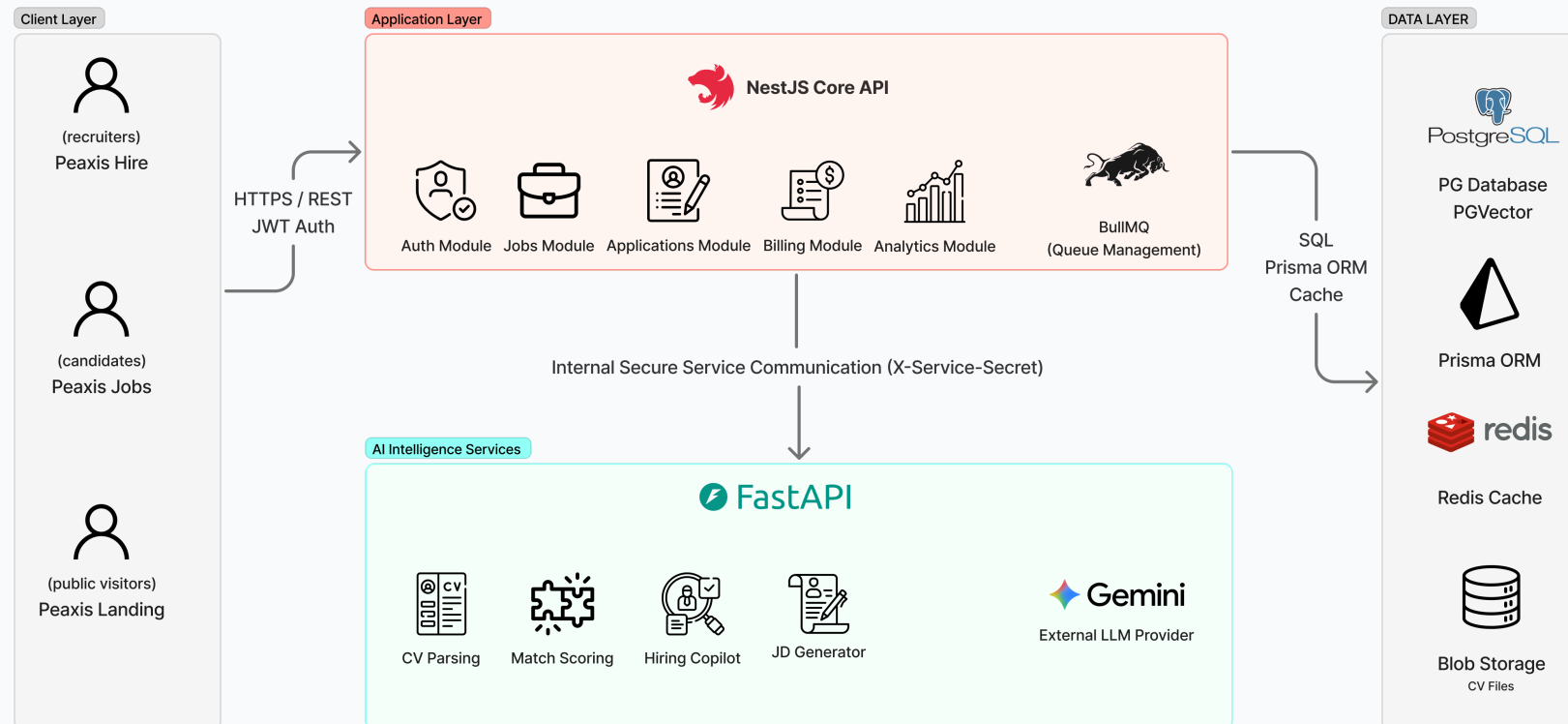
Application Progress

Discover → apply → follow progress

7 ARCHITECTURE & TECHNOLOGIES

Logical Architecture

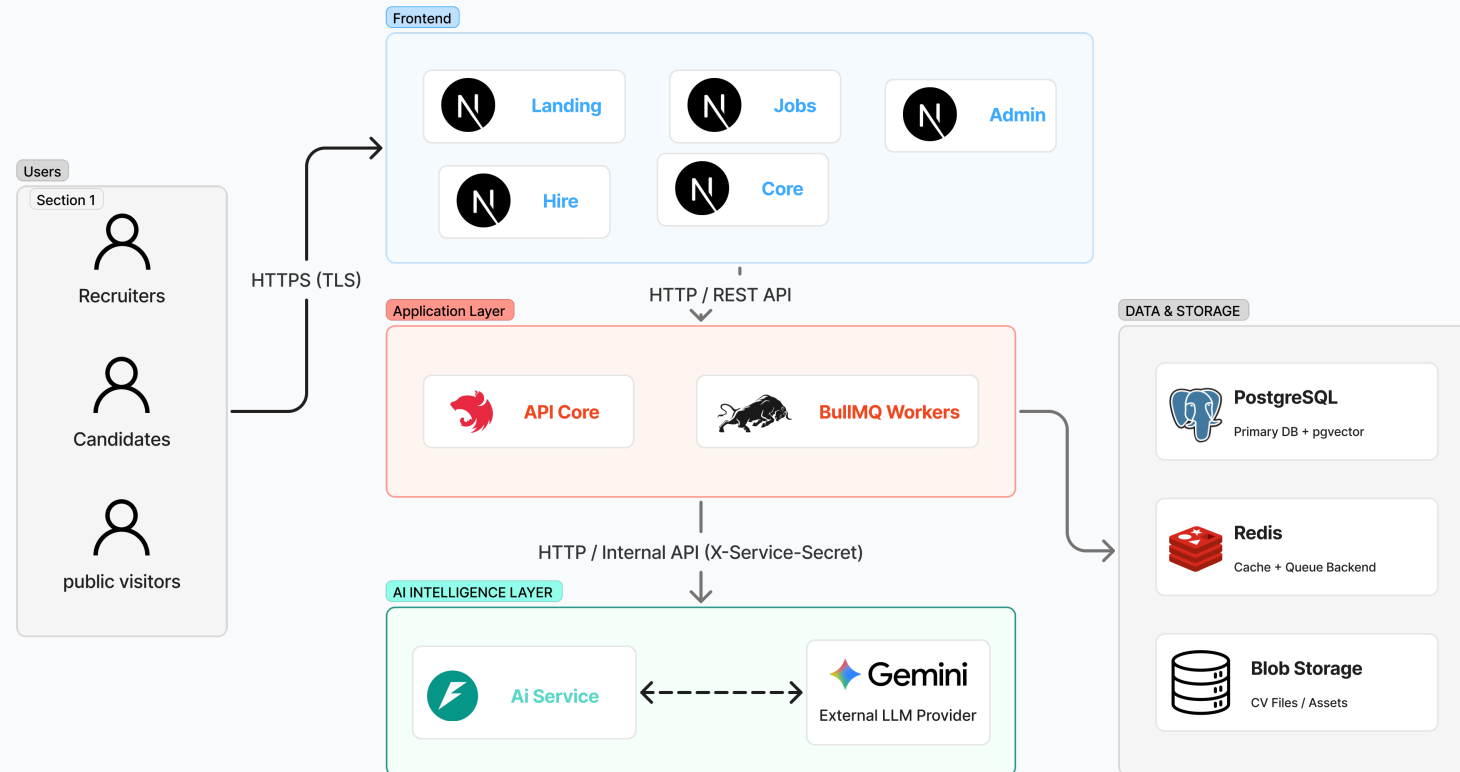
Client, platform, AI, and data boundaries. Gemini or Azure OpenAI is configured at startup.



7 ARCHITECTURE & TECHNOLOGIES

Physical Architecture

Deployment topology for web apps, platform services, AI inference, and storage. Gemini or Azure OpenAI is configured at startup.



8 ENGINEERING DEEP DIVE

AI Runtime Architecture

NestJS orchestrates, FastAPI infers, and PostgreSQL is the source of truth.



Synchronous request path

Validate, authorize, persist business state, and return a processing response.
Short generation or extraction helpers can call FastAPI directly.

Asynchronous worker path

Call FastAPI, persist parse/embedding/assessment outputs, and update the durable work and application state.

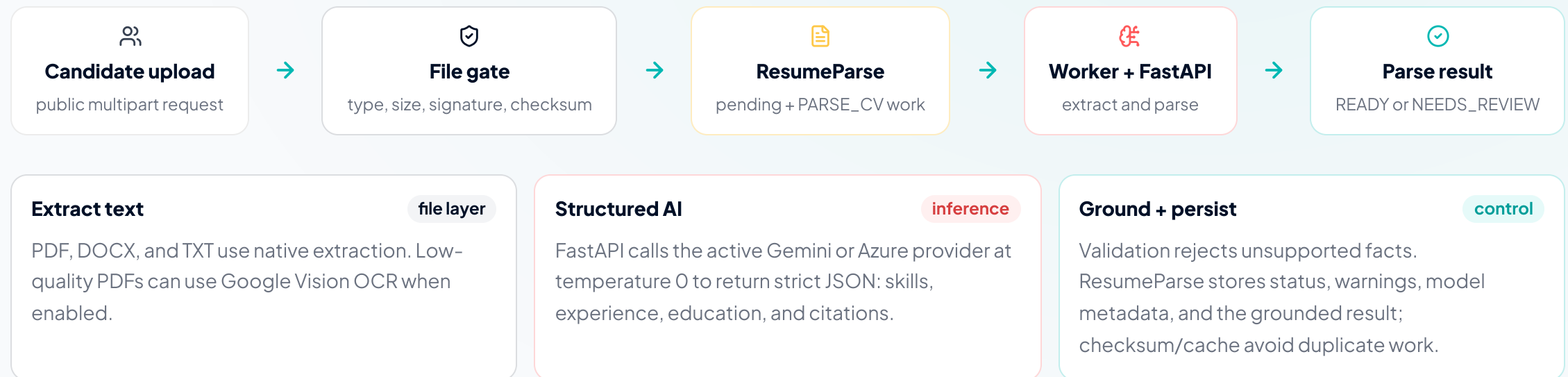
Return path: the worker writes results to PostgreSQL; Redis is used for queues, locks, heartbeats, and caches—not authoritative AI state.

8 ENGINEERING DEEP DIVE

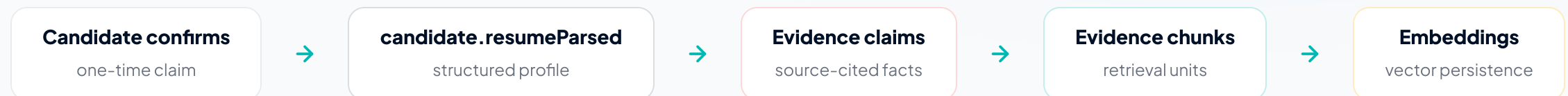
CV Parsing Pipeline

A worker turns a validated CV into grounded, structured evidence — without blocking the candidate.

1. INTAKE AND DURABLE PARSING



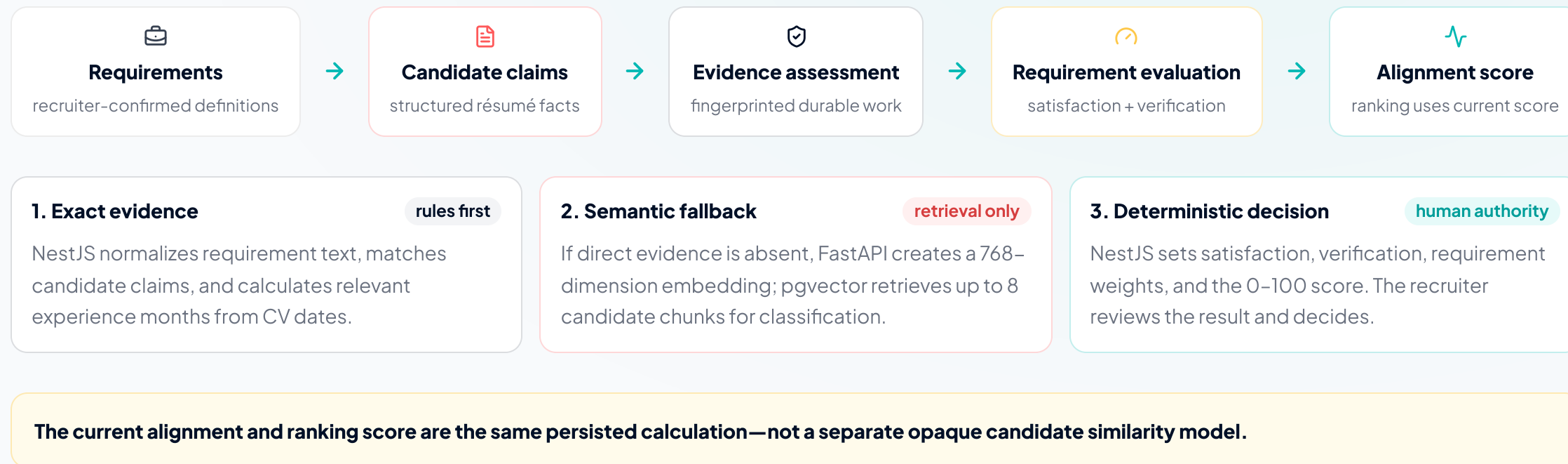
2. CANDIDATE CONFIRMATION AND RETRIEVAL PREPARATION



8 ENGINEERING DEEP DIVE

Evidence-Based Matching Engine

Exact rules first; pgvector retrieval only when needed; the backend always owns the final decision.



How Matching Is Computed

A deterministic evidence algorithm; vectors and LLMs only retrieve or classify citations.

1. Find candidate evidence

Canonical token and simple stem matching compare each requirement with source-cited claims. The fallback searches displayable résumé summary, headline, experience, and project text. Experience requirements merge overlapping date intervals into relevant months.

2. Use vectors only when needed

For unresolved COMPETENCY, RESPONSIBILITY, or TOOL_SYSTEM requirements, FastAPI embeds the requirement. pgvector retrieves only this candidate's chunks at distance ≤ 0.55 (maximum 8). The classifier sees at most 12 claims plus retrieved chunks and may cite only one supplied source.

3. Apply deterministic requirement states

No evidence, mandatory
REQUIRES_VERIFICATION

Evidence but years unmet
PARTIALLY_SATISFIED

No evidence, preferred
UNKNOWN

Evidence + policy satisfied
SATISFIED

4. Persist explainable results

An **EvidenceAssessment** stores input snapshots and the score. Each **RequirementEvaluation** retains satisfaction, constraints, and positive cited evidence.

CURRENT GENERAL: V1 SCORE

$$\text{score} = \text{round} \left(\frac{\sum (\text{weight} \times \text{state value} \times \text{evidence credit})}{\sum (\text{weight})} \right)$$

State value: SATISFIED = 100, PARTIALLY_SATISFIED = 60, all other states = 0. The current ranking score equals this alignment score; no separate opaque cosine ranking is used.

required 0.55

experience 0.25

preferred 0.10

TRANSFERABLE × 0.60

AI Models **by Use Case**

Gemini extracts or classifies evidence; deterministic backend rules calculate the final score.

Use case	Technology	Role
CV parsing	Gemini 3.5 Flash	Strict structured résumé JSON
Evidence classification	Gemini 3.5 Flash	Classify supplied citations only
Embeddings	gemini-embedding-001	768D semantic retrieval with pgvector
OCR fallback	Google Vision OCR	Recover text from poor PDFs
Final score	NestJS + PostgreSQL	Deterministic 0–100 calculation

Gemini-only model path

FastAPI isolates inference

LLMs do not rank candidates

8 ENGINEERING DEEP DIVE

Performance Optimizations

Bounded retrieval, cache reuse, and durable work reduce synchronous AI latency.



Redis cache

Bounded parse, embedding, and selected-generation reuse; never the source of truth.



Durable worker

Slow inference happens outside HTTP requests with task-specific retry policies.



Hybrid job search

PostgreSQL full-text search merges with pgvector semantic results.



Chunk retrieval

Evidence retrieval is constrained to the candidate and returns only a small cited set.



Provider retries

The inference client retries transient provider/network failures.



Graceful search

If query embedding fails, public job search returns keyword results with degraded status.

Known limit: pgvector is enabled, but the baseline migration does not create an ANN vector index.

Technical Challenges & Solutions

Safeguards and limitations remain explicit for technical review.

Challenge	Solution	Outcome
AI latency	Durable work + worker retries	Responsive request path
CV reliability	File gates + quality gate + optional OCR	READY or NEEDS_REVIEW state
Grounding	Citation-bound evidence classification	No free-form evidence invention
Explainability	Snapshots, evaluations, and citations	Auditable assessment trail
Provider outage	Retries and explicit failure states	Recoverable, observable processing
Known access gap	Candidate recommendation visibility needs remediation	Documented before production

Roadmap & Future Work

Next: harden the delivered system before widening AI capability.

NEAR-TERM

Access control

Enforce active membership for all AI-generated content and correct recommendation visibility.

Atomic delivery

Commit application and assessment work atomically with a transactional outbox pattern.

Reviewer policy

Make overrides affect an explicit versioned assessment policy and explainability.

MID-TERM

FinOps coverage

Route every chargeable AI call through operations and credit ledger reconciliation.

Prompt controls

Centralize prompt construction and test every user-controlled input surface.

Vector lifecycle

Add stale-chunk cleanup and measure before introducing an ANN vector index.

LONG-TERM VISION

Governed expansion

Expose only supported FastAPI helpers through tenant-authorized NestJS contracts.

Data governance

Define provider retention, PII classification, consent, and operational redaction.

10 RESULTS & CONCLUSION

Results & Conclusion



Engineering Achievements

- Full-stack recruitment platform



Architectural Achievements

- Controlled AI architecture



Lessons Learned

- Human authority stays central



Evidence, not opaque ranking.

Thank you Questions?

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